

Correct words (Fig. 12) exhibit a pronounced cusp: KL divergence remains low during the early exploratory regime, then rises sharply at the transition (epochs 230–250) as regular usage patterns emerge. Post-transition, KL divergence remains elevated, reflecting that correct-word distributions stabilize in a structured, non-Poisson regime. The cusp indicates the moment when the model transitions from sparse, random correct-word usage to systematic, sub-Poisson regularity. Notably, KL divergence captures distributional features beyond variance: even when dispersion temporarily returns to $D \approx 1$ during the exploratory phase, KL divergence remains elevated, indicating that the distribution shape differs from true Poisson randomness.

Incorrect words (Fig. 13) display the opposite pattern: KL divergence is initially elevated (reflecting the sub-Poisson clustering of repetitive fragments), then converges toward zero as errors become sparse and independent. Post-transition, incorrect words approach the Poisson baseline, consistent with their role as rare, uncorrelated residual noise. The convergence of incorrect-word KL toward zero coincides precisely with the collapse of the incorrect vocabulary and the cusp in correct-word KL.

This divergence–convergence duality occurs in the same critical epoch band (230–250) identified by dispersion and word-length metrics. The fact that KL divergence and dispersion exhibit synchronized cusps—yet measure different statistical properties (full distributional form versus variance structure)—demonstrates that the transition is captured by multiple independent probes. Their simultaneous reorganization provides converging evidence for a genuine phase-transition-like event rather than a metric artifact.

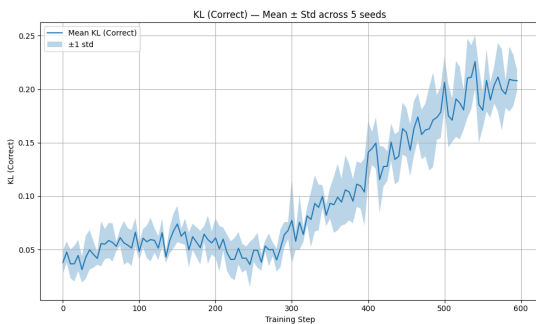


FIGURE 12. KL divergence for correct words to a fitted Poisson baseline across epochs (mean $\pm$ s.d.). A pronounced cusp appears at epochs 230–250 as the model transitions from sparse, near-Poisson usage to structured, sub-Poisson regularity. Post-transition, KL remains elevated, reflecting stable non-Poisson structure.

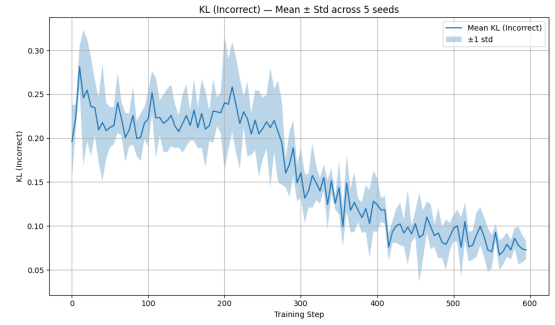


FIGURE 13. KL divergence for incorrect words to a fitted Poisson baseline across epochs (mean $\pm$ s.d.). Initially elevated due to sub-Poisson clustering of fragments, KL converges toward zero as errors become sparse, independent residual noise. The convergence coincides with the correct-word cusp (Fig. 12).

G. INDEX OF DISPERSION (FANO FACTOR)

The index of dispersion ($D = \sigma^2 / \mu$) acts as our primary order parameter, quantifying whether word occurrences follow Poissonian randomness ($D=1$), sub-Poissonian regularity ($D<1$), or super-Poissonian clustering ($D>1$). We analyze correct and incorrect words separately to reveal the regime flip that defines the transition.

Correct words (Fig. 14) begin near Poisson ($D \approx 1$) during early training, reflecting sparse, scattered usage. At the transition (epochs 230–250), dispersion exhibits a temporary dip toward $D \approx 1$ before stabilizing below unity ($D < 1$) post-transition. This sub-Poisson regime indicates that correct words are generated with regular, evenly-spaced patterns rather than random bursts. The temporary return to Poisson during the transition suggests a brief exploratory phase where the model reorganizes its word-generation strategy before committing to structured usage. This transient precisely matches the KL divergence cusp and the histogram flip shown in the Methods section.

Incorrect words (Fig. 15) display the opposite trajectory: they begin in a sub-Poisson regime ($D < 1$) dominated by repetitive short fragments (e.g., repeated “th” or “yo”), then transition toward Poisson ($D \approx 1$) as errors become sparse and independent. Post-transition, incorrect-word dispersion hovers near unity, consistent with random, uncorrelated residual noise. The sub-Poisson $\rightarrow$ Poisson shift for incorrect words mirrors the near-Poisson $\rightarrow$ sub-Poisson shift for correct words, revealing a coordinated reorganization.

The coordination between dispersion and KL divergence is particularly revealing. While dispersion directly measures variance relative to the mean, KL divergence captures the full shape of the distribution. During the transition, both metrics show cusps, but they detect slightly different features: dispersion tracks the regime change (Poisson $\rightarrow$ sub-Poisson for correct words), while KL tracks the magnitude of structural deviation. The fact that both exhibit synchronized discontinuities in the same narrow epoch window—without being mathematically redundant—provides robust evidence that the transition reflects a genuine internal

reorganization rather than an artifact of a single metric.

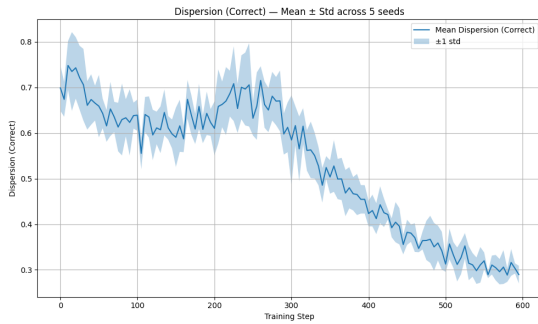


FIGURE 14. Index of Dispersion for correct words ($D = \sigma^2 / \mu$) across epochs (mean $\pm$ s.d.). Begins near Poisson ($D \approx 1$), exhibits a temporary reversion to Poisson at the transition (epochs 230–250), then stabilizes in a sub-Poisson regime ($D < 1$), indicating regular, structured usage.

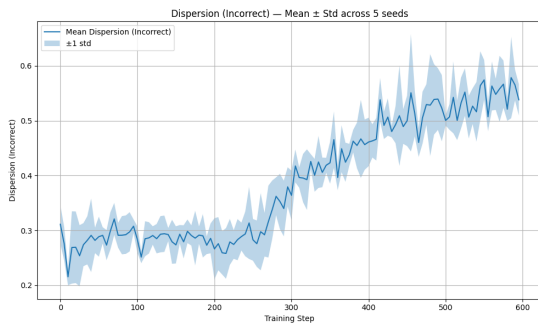


FIGURE 15. Index of Dispersion for incorrect words ($D = \sigma^2 / \mu$) across epochs (mean $\pm$ s.d.). Begins in a sub-Poisson regime ($D < 1$) reflecting repetitive fragments, then transitions toward Poisson ($D \approx 1$) as errors become sparse and independent. The shift mirrors the correct-word transition (Fig. 14).

H. SUMMARY

Across all five seeds, the training trajectory divides naturally into three regimes—before, during, and after a discontinuity centered around steps 230–250.

Before the discontinuity (steps $\lesssim 230$), the model remains in an exploratory regime. Generated text consists mostly of short, fragmented tokens; both dispersion and KL divergence indicate near-Poisson behavior for correct words ($D \approx 1$, low KL); and the incorrect vocabulary slowly expands. Correct words appear sporadically and are typically one or two characters long, indicating limited compositional structure.

During the discontinuity (steps ~ 230 –250), every external metric reorganizes simultaneously. Both dispersion and KL divergence exhibit sharp, synchronized cusps—dispersion temporarily returns to Poisson while KL divergence peaks, indicating maximum distributional deviation. The incorrect vocabulary reaches its maximum diversity, and the model temporarily produces more errors as it experiments with new token combinations. Average correct-word length rises sharply in an S-shaped jump from $\sim 1.5 \rightarrow 2.5$ characters, and prefix tracking (e.g., for “you”) shows isolated fragments fusing into complete, stable words.

This brief phase marks a concentrated restructuring of internal representations, where exploratory variability and combinatorial mixing peak.

After the discontinuity (steps $\gtrsim 250$), exploratory variability subsides. The incorrect vocabulary collapses while the correct vocabulary grows steadily; dispersion drops below unity ($D < 1$) and stabilizes, while KL divergence for correct words remains elevated (reflecting structured, non-Poisson regularity) and KL for incorrect words approaches zero (sparse, independent errors). Generated outputs become longer, coherent, and linguistically stable, reflecting consolidation of learned structure.

Crucially, the synchronization across multiple independent metrics—dispersion (variance structure), KL divergence (full distributional form), word length (linguistic complexity), and vocabulary dynamics (compositional reorganization)—provides converging evidence that this is a genuine phase-transition-like reorganization. No single metric alone would be as convincing; their collective cusp in the same narrow epoch window rules out metric artifacts and supports the interpretation of an abrupt internal restructuring.

V. DISCUSSION

A. POISSON-CENTERED DIAGNOSTICS AND DETECTION IN LINEAR TRAINING SPACE

Our study addresses three central questions about phase transitions in language models. The second question—whether such transitions can be detected directly in linear training space rather than only after log rescaling—is answered affirmatively through our Poisson-centered diagnostic framework. By tracking both dispersion and KL divergence alongside vocabulary statistics, we observe synchronized cusp-like discontinuities that occur directly along the raw epoch axis, without logarithmic transformation of training steps or compute. This demonstrates that emergent reorganizations are intrinsic properties of the learning process, not artifacts of visualization conventions or metric rescaling.

Before the transition, incorrect words display variance suppression ($D < 1$), forming narrow sub-Poisson distributions dominated by repetitive fragments such as “th” or “yo.” Correct words appear sparsely and follow near-Poisson randomness, with low KL divergence from the baseline. At the transition, all metrics converge within a narrow critical band: correct-word dispersion temporarily approaches Poisson ($D \approx 1$) while KL divergence peaks sharply, indicating that although the variance ratio returns to unity, the full distribution shape deviates maximally from true Poisson. This divergence between dispersion and KL during the transition reveals complementary information: dispersion tracks the variance regime, while KL captures the full distributional structure. The fact that both exhibit cusps—but measure different statistical properties—strengthens the case for genuine reorganization rather than a metric artifact.

Afterward, the incorrect vocabulary collapses while correct-word usage stabilizes in a sub-Poisson regime (low D , elevated KL reflecting structured regularity), and average word length increases. This coordinated Poisson $\rightarrow$ sub-Poisson swap across multiple independent probes provides a clear statistical signature of the transition, directly answering our second research question: phase-transition-like discontinuities can be detected in linear training space through Poissonian diagnostics.

B. EARLY LEXICAL REORGANIZATION AS THE TRANSITION MECHANISM

The observed transition is not a gradual refinement of accuracy but a discrete reorganization in the model's word-generation strategy. Evidence across multiple linguistic indicators reveals that the transition is driven by the abrupt emergence of longer, coherent words: (i) the mean correct-word length shows an S-shaped rise from ~ 1.5 to ~ 2.5 characters, representing a jump in linguistic complexity; (ii) prefix tracking of the word "you" reveals consolidation, where single-letter (y) and two-letter (yo) fragments merge into the stable three-letter form (you) precisely at the transition epoch; and (iii) the unique incorrect vocabulary, which expands during fragment proliferation, declines sharply as fragments are absorbed into valid words.

These convergent signatures reveal a qualitative reorganization from fragmentary to compositional word formation. Crucially, this shift occurs early in training—around epochs 230–250—well before training or validation loss stabilizes. The model achieves lexical coherence not through late-stage fine-tuning but through an abrupt mid-training reorganization. This directly answers the third research question: phase-transition-like reorganizations can emerge at early stages of training, marking the onset of lexical coherence rather than late-stage convergence. The synchronized timing across dispersion cusps, KL divergence peaks, vocabulary dynamics, and word-length jumps confirms that these changes reflect a single, coordinated restructuring event.

C. TEMPORARY DEGRADATION AS EVIDENCE OF BARRIER-CROSSING DYNAMICS

A particularly striking feature of the transition—and one that distinguishes genuine phase-transition-like reorganization from gradual learning—is that multiple metrics indicate temporary performance degradation before improvement. The incorrect vocabulary reaches its maximum diversity at step 250 (Fig. 9), meaning the model generates more errors during the transition than in the preceding 100 epochs. Simultaneously, correct-word dispersion reverts toward Poisson ($D \approx 1$) from its earlier sub-Poisson tendency, indicating that the weak regularity established early in training is temporarily disrupted. KL divergence also peaks sharply during this window, confirming maximum distributional instability. The model does not improve monotonically; it briefly becomes worse before reorganizing into a more coherent state.

This "destabilization before consolidation" pattern is agnostic of first-order phase transitions in statistical mechanics, where systems must overcome nucleation barriers to transition between competing stable states. In the framework developed by Rubin, Seroussi, and Ringel for grokking, such behavior arises when an effective potential has two minima—one corresponding to memorization and another to generalization—separated by a barrier. The system must temporarily increase its effective "energy" (disorder, entropy, error rate) to escape the memorization basin and reach the generalization basin. During the barrier-crossing phase, the system exhibits mixed-phase behavior: neither fully in the old regime nor fully in the new, leading to heightened variability and apparent regression.

Our linguistic data provide empirical evidence for this mechanism in a character-level language model. Fragment proliferation—the temporary surge in incorrect vocabulary—is not wasted computation but a necessary exploratory phase. The model must generate and test many erroneous combinations ("th," "yo," "ou," etc.) before discovering which fragments compose into valid words. This exploration is visible as increased error diversity and loss of early regularity. Once sufficient fragments have been tested, the system consolidates: incorrect vocabulary collapses as fragments are absorbed into multi-character words (e.g., "y" + "o" $\rightarrow$ "yo," then "yo" + "u" $\rightarrow$ "you"), and correct-word usage stabilizes into sub-Poisson regularity. The transition is abrupt not because learning suddenly accelerates, but because the system commits to a compositional strategy after exhausting fragmentary exploration.

This finding directly parallels grokking, where models trained on algorithmic tasks exhibit prolonged plateaus in generalization accuracy—sometimes hundreds of thousands of steps—before sudden jumps to near-perfect performance. Power et al. and Agarwal et al. documented that during these plateaus, internal representations are reorganizing even though external accuracy remains flat or fluctuates. Our contribution extends this phenomenon to linguistic tasks and provides statistical signatures—vocabulary proliferation, dispersion reversion, KL divergence peaks—that predict impending reorganization. Unlike grokking studies that focus on memorization-to-generalization transitions in supervised settings, we observe exploration-to-consolidation transitions in unsupervised next-token prediction, suggesting that barrier-crossing dynamics are a general feature of neural training.

The broader implication is that stagnation or temporary degradation during training may signal impending breakthrough rather than failure. Standard training practices often interpret validation plateaus or upticks in error rate as signs to stop training, adjust hyperparameters, or discard the model. Our findings suggest an alternative interpretation: such periods may reflect productive exploration necessary for qualitative reorganization. A model generating more errors at step 250 than step 200 is not necessarily failing—it may be sampling the combinatorial space required to